

DATA ANALYTICS HANDBOOK

The Ultimate Job Ready Data Analytics Resource



CodeWithHarry

What is Data Analytics?

Data analytics means using data to find useful insights that help you make better decisions.

Think of it like this:

- **You collect data:** Collect data like sales numbers, website traffic, survey responses, sensor readings, app usage, etc
- **You clean and organize the data:** Remove errors, fill missing values, standardize formats.
- **You analyze the data:** Use formulas, statistics, tools, and visualizations to understand patterns.
- **You interpret the results:** Convert numbers into actions.
- **You make decisions:** Answer questions like:
 - Which product sells best?
 - Which marketing campaign is working?
 - Where your business is losing money?
 - What customers really want?

Everyday examples

- Netflix analyzes what you watch to recommend the next movie.
- Amazon analyzes your purchases to show relevant products.
- A business analyzes sales to decide how much stock to order.
- A YouTuber analyzes analytics to see which videos perform best.

What data analysts typically do

- Work with Excel, SQL, Python, and dashboards
- Create reports, charts, and visualizations
- Solve business questions using data
- Present insights clearly so teams can act on them

In one line:

Data analytics is the science of turning raw data into meaningful information that drives smarter decisions.

The Data Analytics Workflow

The data analytics lifecycle describes how raw data becomes real business decisions. There are **6 core stages**:

1. Collect

We gather relevant data from different sources.

Examples of data sources

- Excel or CSV files
- SQL Databases
- APIs (e.g., Twitter, YouTube)
- Sensors or CRM systems
- Surveys or web analytics (Google Analytics)

Key points

- Collect only what is needed
 - Understand data ownership and privacy
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2. Clean

Raw data always contains errors. We fix issues to make it ready for analysis.

Common cleaning tasks

- Removing duplicates
- Fixing inconsistent formats (dates, currency, categories)
- Handling missing data
- Standardizing units (Kg vs. g, km vs. miles)

Why it matters

Analysts spend 60 to 80 percent of their time cleaning data. Cleaner data = better insights.

3. Analyze

This is where we start answering business questions using formulas, statistics, or Python/SQL queries.

What we do

- Identify patterns and trends
- Do aggregations (sum, average, counts)
- Compare categories
- Correlation analysis
- Build initial models if needed

Questions we answer

- What happened? (Descriptive)
 - Why did it happen? (Diagnostic)
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4. Visualize

We convert numbers into charts so teams can understand insights quickly.

Tools used

- Excel charts
- Power BI/Tableau dashboards
- Matplotlib/Seaborn charts

Good visuals show

- Trends
- Comparisons

- Segmentation
 - Outliers
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5. Report

We communicate insights in a clear and actionable way, typically to non-technical stakeholders.

This includes

- Summary dashboards
- Slide decks
- Business recommendations
- Email updates

Important skill **Storytelling**: Present insights like a narrative, not just numbers.

6. Decision Making

The final business decisions are taken based on what the data shows.

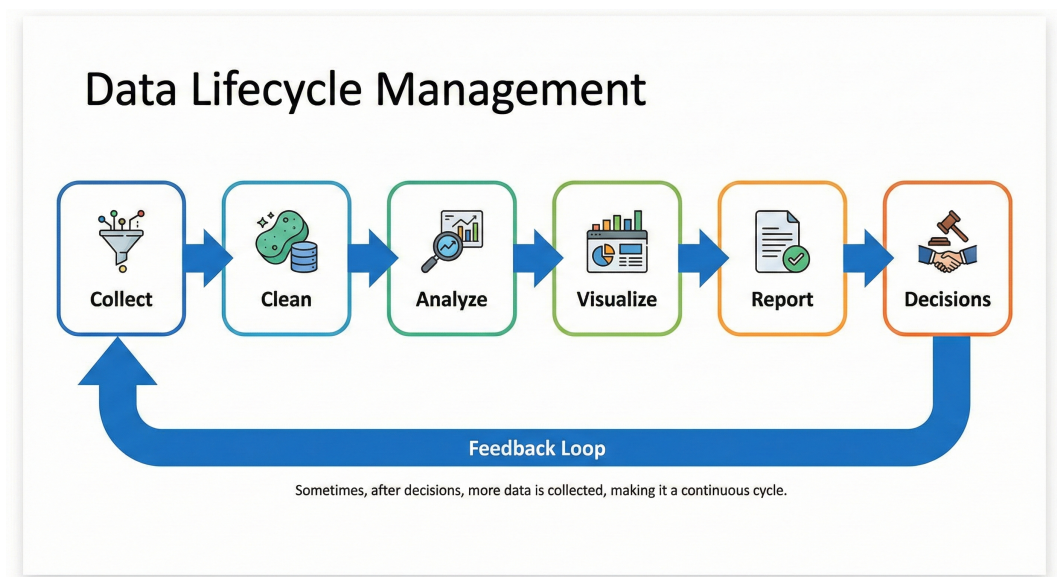
Examples

- Improve product pricing
 - Change marketing strategy
 - Fix customer dissatisfaction points
 - Remove low-performing features
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Real Example Workflow: Online Store Sales Analysis

Business Goal: Improve monthly revenue of an e-commerce store.

Step	What happens	Example
Collect	Sales data from SQL database + customer data from CRM	Orders table, Users table
Clean	Remove duplicate orders, fix date formats, fill missing values	Correct wrong prices
Analyze	Find top-selling products, customer repeat rate	Calculation of revenue by category
Visualize	Charts showing sales by category, month, customer type	Line chart + bar chart
Report	Create dashboard + summary recommendations	"Category A drives 45 percent revenue"
Decision	Increase stock + run ads on best products	Revenue improves next month



Data Analytics Lifecycle

Tools used by a Data Analyst

Data analysts use a variety of tools to collect, clean, analyze, visualize, and report data. Here are some of the most common ones:

- **Data Collection:** SQL, Google Analytics, APIs, Web Scraping
- **Data Cleaning:** Python & Python libraries like Pandas, Numpy, etc.
- **Data Analysis:** Excel, SQL, Python (NumPy, Pandas) etc.
- **Data Visualization:** Tableau, Power BI, Matplotlib, Seaborn
- **Reporting:** Excel, PowerPoint, etc.

Each tool has its strengths and is chosen based on the specific needs of the project.

Mini Project: The Profitable Car Showroom

A Mahindra showroom manager wants to know which car models are bringing higher profit and which models need better marketing or offers.

Dataset

Below are the car models sold in the showroom along with units sold per month and profit per car.

Model	Units Sold per Month	Profit per Car (₹)
Bolero	25	1,50,000
Scorpio	40	1,90,000
Thar	30	2,20,000
XUV700	45	2,50,000
XUV300	50	1,30,000

Task

Add a new column → Monthly Profit Formula: Units Sold × Profit per Car

By quick mental math or calculator:

Model	Units Sold	Profit per Car	Monthly Profit
Bolero	25	1,50,000	₹37,50,000
Scorpio	40	1,90,000	₹76,00,000
Thar	30	2,20,000	₹66,00,000

Model	Units Sold	Profit per Car	Monthly Profit
XUV700	45	2,50,000	₹1,12,50,000
XUV300	50	1,30,000	₹65,00,000

Basic Insights

- XUV700 generates the highest total profit for the showroom.
- Even though XUV300 sells the most units, its lower profit per car gives it less total profit.

Message for Students

Data analytics is not about guessing. It is about using data to help the business take decisions such as:

- Which car to promote more
- Which car needs more marketing
- Which car generates maximum revenue

This is what a data analyst does. We use data to tell the business which product brings money and which product needs help. This is only the start. Soon you will learn Excel, SQL, AI and Python to work with much bigger and complex data.